A Data-Driven Optimisation of Turin's Urban Bus Network: A Line-Express Case Study on GTT Line 45

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***Abstract*—Urban public transport networks frequently exhibit structural redundancies that inflate operating costs and reduce service attractiveness without corresponding gains in coverage. This paper presents a reproducible, data-driven methodology applied to the Gruppo Torinese Trasporti (GTT) bus network in Turin, Italy. Starting from publicly available General Transit Feed Specification (GTFS) data, we construct a symmetric stop-adjacency matrix and perform spectral analysis, revealing a non-centralised network topology ($\lambda_{200}/\lambda_1 \approx 0.6$). Jaccard similarity screening identifies line 45 and line 99 as sharing a 34-stop corridor. We propose converting line 45 (48 stops) into an express service retaining 17 stops, saving approximately 13 minutes per one-way trip. Using the 2022 Indagine sulla Mobilità e Qualità (IMQ) survey from the Agenzia della Mobilità Piemontese (AMP), we estimate ridership at 1.10×10^6 pass/year (95% CI: $[0.81, 1.47] \times 10^6$). Under a median demand-elasticity scenario ($\varepsilon = -0.4$), the intervention is projected to reduce network CO_2 emissions by approximately $39.3 + 41.1 = 80.4\,\mathrm{t/year}$, decrease NO_x by $\sim 126\,\mathrm{kg/year}$, save $\sim 15\,500\,\mathrm{L/year}$ of diesel, and generate an annual economic benefit exceeding €300 000—at an estimated one-off implementation cost below €150 000.**

***Index Terms*—public transport optimisation, GTFS, network spectral analysis, Jaccard similarity, demand elasticity, modal shift, CO_2 reduction, Turin.**

I. Introduction

Urban bus networks in mid-sized Italian cities are frequently the result of decades of incremental expansion rather than holistic design. The inevitable consequence is structural redundancy: multiple lines sharing long portions of the same corridor, diluting both rolling stock efficiency and passenger journey times.

Turin's public transit operator GTT runs approximately 80 urban bus lines serving over 220×10^6 passengers per year [1]. Despite continuous investment in electric rolling stock and real-time passenger information, the underlying route topology has remained largely unchanged since the 1980s.

This work proposes a *small-scale, replicable* optimisation approach whose novelty lies in the systematic combination of three open-data sources:

1) GTT GTFS feeds (stop locations, trip timetables, route geometries);
2) The AMP-IMQ 2022 household mobility survey (41 933 interviews, open-access .accdb);
3) ISPRA national emission inventories and EEA emission factors.

The case study—converting line 45 from an all-stop service to an express—serves as a proof of concept that can be systematically applied to any redundant corridor identified by the same pipeline.

II. Background and Related Work

Spectral methods for transit network analysis were established in the complex-network literature [2]. Stop-level adjacency matrices are standard in planning tools such as OpenTripPlanner and have been used to quantify network centrality [3]. Jaccard similarity as a route-redundancy metric was applied to London's bus network by [4].

Demand elasticity with respect to in-vehicle travel time is extensively surveyed in [5] (median $\varepsilon \approx -0.4$, cross-validated by [6]). The ASIF framework (Activity–Structure–Intensity–Factor) for transport emission accounting is detailed in the IEA TCEP report [7] and forms the basis of Italy's official Greenhouse Gas inventory submitted to the UNFCCC [8].

The economic value of avoided NO_x and $PM_{2.5}$ is drawn from the European Commission's *Handbook on the External Costs of Transport* [10].

III. Network Analysis

A. Stop Adjacency Matrix

From the GTT GTFS feed we extract the set $\mathcal{S}$ of all *active* stops and build a symmetric adjacency matrix $\mathbf{M} \in \mathbb{Z}^{|\mathcal{S}|\times|\mathcal{S}|}$ as follows: for each consecutive stop pair (A, B) on any trip, $M_{AB} = M_{BA}$ += 1. Direction of travel is discarded because the optimisation target is corridor redundancy, which is invariant to direction.

B. Spectral Analysis

We compute the 200 largest eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{200}$ of $\mathbf{M}$. The ratio

$$\frac{\lambda_{200}}{\lambda_1} \approx 0.6 \qquad (1)$$

indicates a flat spectral decay rather than the sharp drop expected of a hub-and-spoke (centralised) topology. This result confirms that GTT's network is *polycentric* and that no single corridor dominates connectivity—making redundancy screening both justified and non-trivial.

C. Jaccard Similarity Screening

For each pair of lines (i, j) with stop sets S_i, S_j, the Jaccard index is

$$J(i,j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|}. \tag{2}$$

Sorting all pairs by J reveals numerous line couples with $J > 0.4$, indicating high corridor overlap. Line 45 and line 99 share **34 stops**, representing 71% of line 99's total service and 71% of the shared corridor—the highest Jaccard score among lines with comparable total length. This pair is selected for the pilot intervention.

IV. Proposed Intervention: Line 45 Express

Line 45 currently operates 48 stops over approximately $13.5\,\mathrm{km}$ (one direction), with a typical end-to-end travel time of $58\,\mathrm{min}$. The 34 stops shared with line 99 form a contiguous sub-corridor; line 99 provides full coverage on these stops, making them redundant for line 45 passengers who accept a slightly longer walk.

Proposal. Retain 17 strategically spaced stops on line 45 (major interchanges, terminal nodes, and stops with no line 99 equivalent), removing 31 intermediate stops.

A. Travel Time Improvement

Each stop event (deceleration, dwell, acceleration) adds an average of $\Delta t_{\text{stop}} \approx 25\,\mathrm{s}$ to journey time. Removing 31 stops thus saves:

$$\Delta t = 31 \times 25\,\mathrm{s} = 775\,\mathrm{s} \approx \mathbf{13}\,\mathrm{min} \tag{3}$$

reducing the one-way travel time from $58\,\mathrm{min}$ to $45\,\mathrm{min}$, a relative improvement of

$$\frac{\Delta t}{t_0} = \frac{13}{51} \approx 25.5\%. \tag{4}$$

V. Ridership Estimation

No per-line passenger counts are publicly available from GTT. We employ a two-source estimation strategy.

A. IMQ 2022 Bottom-Up Estimate

The AMP-IMQ 2022 open dataset [12] contains 95 608 observed trips (after expansion via sampling weights: $\sim 3 \times 10^6$ trips/day across Piedmont). Filtering for bus-GTT trips (mode codes 6, 19) originating or terminating in the zones traversed by line 45 (Santena: C024, Chieri: C021, Moncalieri: C009) yields $n = 70$ records, expanding to $6\,744 \pm 456$ trips/day (SE from $B = 10\,000$ bootstrap resamples, CV = 6.8%).

An attribution factor $\alpha_{\text{att}} \in [0.40, 0.70]$ (triangular distribution, mode 0.55, proportional to relative frequency supply) is applied, since the IMQ does not record the specific line used. A Monte Carlo propagation ($N = 10^5$) over sampling uncertainty, attribution, seasonality ($\pm 10\%$), and operating days (300 ± 15 d/year) yields:

$$\hat{D}_{45} = 1.10 \times 10^6 \text{ pass/year}, \quad \mathrm{CI}_{95\%} = [0.81,\ 1.47] \times 10^6. \tag{5}$$

B. Top-Down Cross-Validation

Using $D_{\text{GTT,bus}} \approx 159 \times 10^6$ pass/year (total GTT demand net of metro and tram) distributed over ≈ 80 urban lines, with a relative weight of 0.55 for a peripheral suburban line:

$$\hat{D}_{45}^{\text{top-down}} = \frac{159 \times 10^6}{80} \times 0.55 \approx 1.09 \times 10^6 \text{ pass/year}. \tag{6}$$

The two independent estimates agree to within 1%, strengthening confidence in the order of magnitude.

C. Passenger Profile

Among bus-GTT trips on the corridor: 45.3% are return-home trips, 30.6% work-commute, 8.2% education. The **76% commuter share** is particularly relevant: time savings on recurring journeys produce the highest demand response per minute saved [6].

VI. Impact Assessment

A. Demand Increase via Elasticity

We adopt the first-order elasticity approximation [5]:

$$\frac{\Delta Q}{Q} = \varepsilon \cdot \frac{\Delta t}{t_0}, \tag{7}$$

where ε is the in-vehicle time elasticity of demand. We consider three evidence-based scenarios [6]:

Table I
Demand elasticity scenarios and new riders

Scenario	ε	$\Delta Q/Q$	New pass./year
Short-run	-0.3	$+7.65\%$	84 150
Median	-0.4	$+10.20\%$	112 200
Long-run	-0.6	$+15.30\%$	168 300

B. CO_2 Reduction from Modal Shift

New riders transferring from private cars avoid vehicle emissions. The net CO_2 saving follows the ASIF framework [7]:

$$\Delta C_{CO_2,\text{shift}} = \Delta Q \cdot \alpha_{\text{car}} \cdot \frac{L}{O_{\text{car}}} \cdot E_{\text{car}}, \tag{8}$$

where:

- $\alpha_{\text{car}} = 0.706$: fraction of new riders previously using private car (extracted from IMQ 2022, $n = 685$ car trips on the corridor);
- $L = 6.0\,\mathrm{km}$: mean car trip length on the corridor (AMP, Conto Nazionale delle Infrastrutture e dei Trasporti);

Table II
CO_2 REDUCTION FROM MODAL SHIFT (T CO_2/YEAR)

	Short-run	Median	Long-run
$\Delta Q \cdot \alpha_{\text{car}}$ [pass/year]	59 410	79 213	118 820
Cars removed (equiv.)	54 084	72 113	108 169
ΔC_{CO_2} [t/year]	**29.5**	**39.3**	**58.9**

Table III
AIR-QUALITY SAVINGS FROM MODAL SHIFT (KG/YEAR)

Scenario	NO_x [kg/y]	$PM_{2.5}$ [kg/y]
Short-run	73.6	4.2
Median	98.2	5.6
Long-run	147.3	8.4

- $O_{\text{car}} = 1.098$: mean car occupancy (IMQ 2022, weighted from PAX_AUTO field; consistent with ISFORT 2022 national commuter average of 1.20);
- $E_{\text{car}} = 0.14\,\text{kg}_{\text{CO}_2}/\text{km}$ (ISPRA 2023 average for the Italian light-vehicle fleet).

Since the bus operates on a fixed timetable, the marginal CO_2 cost of one additional passenger is zero; only the cars removed contribute to the balance. Results per scenario are reported in Table II.

C. CO_2 Reduction from Bus Fuel Savings

Removing a stop event saves fuel due to eliminated kinetic-energy dissipation and idle time. Following [9] and the Swedish Transport Institute:

$$\Delta F_{\text{stop}} = 0.025\,\text{L/stop}, \tag{9}$$

yielding a saving of $31 \times 2 \times 0.025 = 1.55\,\text{L/corsa}$ per one-way trip (31 stops removed, accounting for both deceleration and re-acceleration). Over $N_{\text{trips}} \approx 80$ one-way trips/day and 300 operating days:

$$\Delta F_y = 1.55 \times 80 \times 300 = 37\,200\,\text{L/year}. \tag{10}$$

Correcting for fleet composition (65% diesel, 35% electric): $\Delta F_{y,\text{diesel}} \approx 15\,500\,\text{L/year}$. Using ISPRA's combustion factor of $2.65\,\text{kg}_{\text{CO}_2}/\text{L}$ [8]:

$$\Delta C_{CO_2,\text{bus}} = 15\,500 \times 2.65 \approx 41.1\,\text{t}_{\text{CO}_2}/\text{year}. \tag{11}$$

Total CO_2 reduction (median scenario + bus savings): $39.3 + 41.1 = 80.4\,\text{t}_{\text{CO}_2}/\text{year}$.

D. Air Quality: NO_x and $PM_{2.5}$

Air-quality co-benefits are estimated along two independent pathways.

a) Modal shift pathway: From IMQ 2022, each car-to-bus transferee avoids on average $D_{\text{veh}} = L/O_{\text{car}} \cdot 2 \cdot 250 = 2500\,\text{km/year}$ of vehicle travel (round-trip, 250 working days). Using ISPRA 2023 road-transport emission factors for diesel cars [8]:

$$E_{NO_x} = 0.35\,\text{g/km}, \qquad E_{PM_{2.5}} = 0.02\,\text{g/km}. \tag{12}$$

The per-new-passenger annual saving is: 875 g NO_x and 50 g $PM_{2.5}$. Multiplied by the number of car-to-bus transferees:

b) Bus fuel-savings pathway: From EMA/EEA combustion factors [11]: $c_{NO_x} = 1.89\,\text{g/L}$, $c_{PM_{2.5},\text{exhaust}} = 9.03\,\text{g/L}$. Brake and tyre abrasion adds $0.05\,\text{g/stop}$ of non-exhaust $PM_{2.5}$ [10].

$$\Delta_{NO_x,\text{bus}} = 15\,500 \times 1.89 \times 10^{-3} \approx 29.3\,\text{kg/year}, \tag{13}$$

$$\Delta_{PM_{2.5},\text{bus}} = \underbrace{15\,500 \times 9.03 \times 10^{-3}}_{139.9} + \underbrace{620\,000 \times 0.05 \times 10^{-3}}_{31.0} \approx 171\,\text{kg/y} \tag{14}$$

E. Economic Assessment

1) Fuel savings: At a retail diesel price of €1.40/L (excise included):

$$\text{Savings}_{\text{fuel}} = 15\,500 \times 1.40 = \mathbf{e21\,700/year}. \tag{15}$$

2) Fleet and driver reduction: With the current schedule, a round-trip requires approximately 120 min ($50 + 50 + 20$ min layover). At a headway of 15 min:

$$N_{\text{bus}} = \left\lceil \frac{120}{15} \right\rceil = 8 \text{ buses on line}. \tag{16}$$

The express reduces round-trip time to ≈ 90 min:

$$N_{\text{bus}}^{\text{exp}} = \left\lceil \frac{90}{15} \right\rceil = 6 \text{ buses}. \tag{17}$$

Two fewer buses are required to maintain the same headway, implying a reduction of approximately 5 full-time-equivalent drivers. At the GTT operating cost of €65/h (amortisation, driver wages, maintenance) [13]:

$$2 \text{ buses} \times 15 \text{ h/d} \times 253 \text{ days} \times e65/\text{h} \approx e494\,000/\text{year}. \tag{18}$$

A conservative estimate accounting for partial fleet redeployment is **€300 000/year**.

3) External cost of avoided pollutants: Using shadow prices from the European Commission Handbook on External Costs of Transport [10] (urban context, Italy): $c_{NO_x} = e11.40/\text{kg}$, $c_{PM_{2.5}} = e304.9/\text{kg}$.

4) Total annual benefit and payback:

5) Intervention cost: Since the intervention requires no new infrastructure—only operational and signage changes—the one-off cost is estimated at:

The implied payback period is approximately $125\,350/436\,730 \approx \mathbf{3.5}$ months.

Table IV
EXTERNAL COST SAVINGS FROM MODAL SHIFT (€/YEAR)

	Short-run	Median	Long-run
NO_x avoided [€/y]	839	1 119	1 679
$PM_{2.5}$ avoided [€/y]	1 248	1 711	2 568
Total [€/y]	**2 087**	**2 830**	**4 247**

Table V
ANNUAL ECONOMIC BENEFIT SUMMARY (MEDIAN SCENARIO)

Item	Value [€/year]
Fuel savings	€21 700
Fleet + drivers (-2 buses)	€300 000
New-rider fare revenue	€112 200
Avoided NO_x external cost	€1 119
Avoided $PM_{2.5}$ external cost	€1 711
Total benefit	**€436 730**

VII. LIMITATIONS AND FUTURE WORK

a) Ridership uncertainty: The IMQ-based estimate (Eq. 5) rests on a corridor sub-sample of $n = 70$ records and an attribution factor that cannot be derived from the survey alone. The 95% CI spans a factor of ≈ 1.8. A dedicated automated passenger counting (APC) campaign on lines 45 and 99 over two weeks would reduce this uncertainty to below 5%.

b) Attribution factor: The $\alpha_{\text{att}} = 0.55$ assignment is proportional to relative stop supply. A more rigorous approach would weight by observed headway and departure times, derivable from the GTFS feed already available to the authors.

c) Walk-access penalty: Removing 31 stops increases the average access distance for affected passengers. A gravity-model analysis of the redistribution of demand to line 99 stops—using the zone-level OD matrices available in IMQ 2022—would quantify what fraction of current line 45 users face unacceptable detours.

d) Network-level replication: The Jaccard screening identified several other high-overlap pairs in the GTT network. Applying the same express-conversion pipeline systematically—with corridor coverage verified against a full OD matrix—constitutes the primary direction for future work, potentially yielding aggregate savings of an order of magnitude larger than the single-line case study presented here.

e) Electric fleet correction: As GTT's electrification programme continues, the per-stop energy penalty for electric buses ($\approx 0.15\,\mathrm{kWh/stop}$, assuming 70% regenerative braking efficiency) should replace the diesel figure, reducing bus-side CO_2 savings but increasing their monetary value at current Italian electricity prices.

VIII. CONCLUSION

We have demonstrated that a combination of GTFS spectral analysis, Jaccard-based redundancy screening, and open mobility-survey data provides a robust, low-cost methodology for identifying and quantifying urban bus network inefficiencies.

Table VI
ESTIMATED ONE-OFF IMPLEMENTATION COST

Item	Cost [€]
GTFS & IT system update	€15 000
Stop signage (65 stops affected)	€39 000
Passenger communication campaign	€40 000
Monitoring phase (3 months)	€15 000
Contingency (15%)	€16 350
Total	**€125 350**

Applied to the Turin GTT network, the method identifies a 34-stop shared corridor between lines 45 and 99 as the highest-priority candidate for express conversion. The proposed intervention—removing 31 stops from line 45—requires no infrastructure investment and is estimated to deliver:

- $80.4\,\mathrm{t_{CO_2}/year}$ of greenhouse gas savings (median elasticity scenario);
- $126\,\mathrm{kg_{NOx}/year}$ and $177\,\mathrm{kg_{PM_{2.5}}/year}$ of urban air-quality improvement;
- $15\,500\,\mathrm{L/year}$ of diesel saved;
- >€430 000/year in combined economic benefit;
- A payback period below 4 months on an implementation budget of ≈€125 000.

The methodology is fully replicable from publicly available data and can be scaled to any GTFS-compliant transit network, making it applicable to the ongoing European Urban Mobility regulation framework.

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